7	(a)			increasing current causes increasing flux (in core) (according to Faraday's law of induction,) induces e.m.f. in coil (from Lenz's law, induced e.m.f.) opposes the increase of current (in circuit)	B1 B1 B1
	(b)			constant maximum current gives constant flux / field induced e.m.f. only when flux (in core/coil) is changing (so, no induced e.m.f.)	B1 B1
	(c)	(i)		$\frac{N_S}{N_P} = \frac{V_S}{V_P}$ r.m.s. voltage of output $= \frac{52}{\sqrt{2}} (= 36.77\ \text{V})$ $N_S = \frac{52/\sqrt{2}}{90} \square 1200 = 490$ (*accept 491 turns*)	 C1 A1
		(ii)		0 ms or 7.5 ms or 15.0 ms or 22.5 ms	A1